

Interleaved Boost PFC with Half Bridge LLC Resonant Converter based EV Battery Charger

Jeevanand P

Department of Electrical and Electronics Engineering
Government College of Engineering Kannur
Kerala, India
jeevanand@gcek.ac.in

Akhil Chacko

Department of Electrical and Electronics Engineering
Government College of Engineering Kannur
Kerala, India
akhilchackoeee@gmail.com

Rajesh M

Department of Electrical and Electronics Engineering
Government Engineering College Wayanad
Kerala, India
mrjesh@gcek.ac.in

Abstract—Electric scooters are attaining demand over conventional ICE scooters due to increases in fuel costs, depletion of fossil fuels, and reduced maintenance in EVs. This paper presents a 1kW two-stage onboard charger consisting of an interleaved boost PFC converter in the primary stage and a half-bridge LLC resonant converter at the secondary stage. The PFC stage maintains a voltage of 310V which is regulated to charge a 60V 86Ah electric scooter battery by the LLC resonant converter. The converter PFC stage is operated in discontinuous conduction mode to achieve natural power factor correction reduce the burden of implementation of complex control circuits and increase the reliability of the converter. The half-bridge topology in the secondary stage reduces the number of switches required without affecting the efficiency of the converter. The secondary stage converter is further operated in zero voltage switching mode to reduce the switching losses. The simulation results show that the system can perform satisfactorily achieving power factor correction and managing the THD level at acceptable limits for EV battery chargers.

Keywords—Interleaved Boost Converter, Power Factor Correction, LLC Resonant Converter

I. INTRODUCTION

There is a new trend in electrification of the automotive industry due to the environmental concerns of internal combustion engine (ICE) based vehicles and advancement in research and development in the EV industry. The chargers for electric vehicles can be categorized into on-board (OBC) and off-board chargers. The latter is implemented in Level II and Level III charging where quick charging is required, also the off-board chargers are of higher power and bulkier compared with the onboard chargers [1]. OBCs are limited to lower power ratings due to constraints in cost and space. They can be classified into unidirectional or bidirectional, isolated or nonisolated, and single-stage or two-stage battery chargers. Typically for lower-power applications unidirectional chargers are preferred. Bidirectional chargers can charge the battery from the grid and supply power back to the grid when required. The single-stage converter is compact and requires less space compared

to its counterpart but the power quality and performance are not ideal for EV battery charging [2]. Unlike EV cars smaller EVs like electric scooters and E-Rikshaws are comprised of small battery packs in the range of 48-60V with OBC of 1 to 3.3kW [3]. A two-stage EV battery charger consists of a PFC converter in the primary stage and a DC-DC converter in its secondary stage. The secondary stage converter is usually a highly efficient and compact converter with the capability of zero voltage switching (ZVS) [4]. It can be an LLC resonant converter or a phase-shifted converter topology. A half-bridge converter topology can be used to reduce the amount of components required and increase the compatibility of the onboard charger [5]. The Boost converter-based converters are widely used PFC converters. The interleaving PFC converters show higher current ripple reduction with the addition of FETs and inductors. The interleaved PFC converter can be with the Diode bridge rectifier (DBR) or without it. Bridgeless topology is comparatively efficient due to the elimination of voltage drop and losses in the DBR [6], while the bridged topologies are simpler and suitable for low-power applications like E Rickshaw and Electric scooter battery chargers.

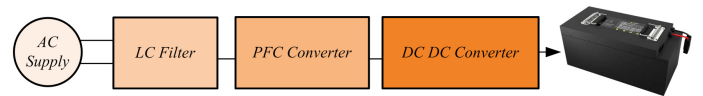


Fig. 1. Block diagram of two stage battery charger

II. SYSTEM CONFIGURATION

The battery charger consists of an interleaved boost PFC converter (IBPFCC) at the primary stage and a half-bridge LLC resonant converter at the secondary stage. The IBPFCC has two switches operated with 180° phase shift between the two. This phase-shifted operation of the switches leads to reduced current ripples at the supply side. The input current

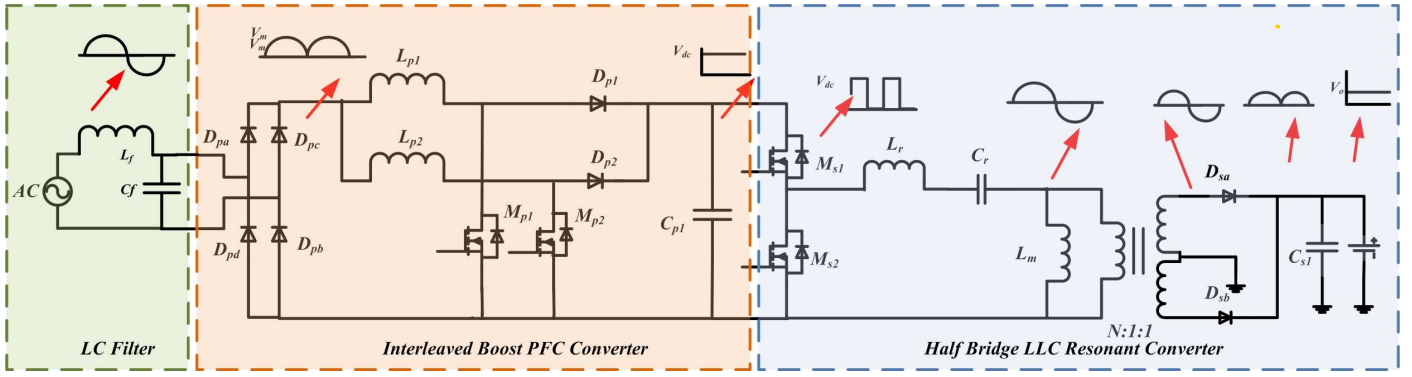


Fig. 2. Circuit diagram of interleaved boost PFC with half bridge LLC resonant converter based battery charger

ripple minimization varies with the duty ratio. The best ripple current reduction is obtained at a 50% duty ratio [7]. The PFC converter is executed in Discontinuous conduction mode (DCM) to achieve the power factor correction naturally without the addition of complex control logic and reducing the stress on the requirement of additional sensors for PFC. The peak current of the switches and diodes in DCM operation will be higher than that of the continuous conduction mode (CCM), so the components with higher RMS current rating are required for DCM operation [8]. The circuit configuration of the converter is given in Fig 2. The secondary stage of the converter is a Half Bridge LLC Resonant converter. The switching frequency and resonant frequency are selected based on the achievement of the ZVS and reducing the complexity in designing the input lowpass LC filter, which is added to reduce the switching ripples reflected at the supply current. The LC filter is designed using the method presented in [9]. To reduce the number of components without compromising efficiency, a half-bridge topology is used for the LLC resonant converter instead of the full bridge topology. The gain of the converter depends on the frequency, the resonant tank, the topology used, and the transformer turns ratio. The primary side of the transformer consists of the resonant tank and the switching stage, provided by the two switches. The secondary of the transformer is linked to a center tap diode rectifier and an output capacitor filter [10].

III. DESIGN OF FRONT END AND BACK END CONVERTERS

The design of the battery charger is done such that the power factor of 0.9-1 is achieved at the supply side while reducing the supply current THD to less than 5% and achieving zero voltage switching at the DC-DC conversion stage. Further, the primary stage converter needs to be operated in DCM.

A. IBPFCC

To operate the converter in DCM the inductance can be designed using the equation

$$L < \frac{V_{indc}(1 - \frac{V_{indc}}{V_{dclink}})}{f_s * \Delta I_{max}} \quad (1)$$

where f_s is the switching frequency, V_{indc} is the rectified supply voltage, ΔI_{max} is the ripple current which is taken

as 0.4% of the load current and V_{out} is DC link voltage. The output capacitor filter removes the voltage ripples and provides a reliable voltage for the secondary stage converter. The current flowing through the output filter capacitor is the sum of the two diode currents.

$$C_{out} = \frac{I}{\pi * f_s * \Delta V_{ripple}} \quad (2)$$

The Duty Ratio is calculated from the following equation of boost converter.

$$\frac{V_0}{V_{in}} = \frac{1}{1 - D} \quad (3)$$

B. LLC

The critical parameters required while designing the converter are the Quality Factor and the Resonant Frequency. The R_{ac} which is the equivalent load that is seen in the primary of the transformer is given by

$$R_{ac} = \frac{8}{\pi^2} \frac{N_p^2}{N_s^2} \frac{V_{out}^2}{P} \quad (4)$$

The Quality factor, Q is selected based on an iterative process beginning with $Q = 0.5$, the lesser the Q , the higher the gain and lesser frequency range of operation, and the higher the Q , the higher the stability. Q is selected as 1.5. The Voltage Gain graph for different Quality Factors chosen in this iterative process is given in Fig 3.

$$Q = \frac{1\sqrt{L_r}}{R_{ac}\sqrt{C_r}} \quad (5)$$

The resonant inductor and capacitor are selected using these equations.

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \quad (6)$$

The higher the m , the lesser the circulating current in the converter. While Designing the transformer the magnetizing inductor can be incorporated in it. Also, the leakage inductance of the transformer can be an added advantage while designing the resonant inductor.

$$m = \frac{L_m + L_r}{L_r} \quad (7)$$

The transformer turns ratio is given by,

$$n = \frac{V_{dclink}}{2 * V_{out}} \quad (8)$$

The center tap diode rectifier is used instead of a full diode bridge rectifier to reduce the voltage drop and size of the converter on the secondary side. ZVS is obtained when the switching frequency is operated near the resonant frequency.

$$F = \frac{f_s}{f_r} \quad (9)$$

The maximum voltage gain of the LLC resonant converter is obtained between f_p and f_r , where

$$f_p = \frac{1}{2\pi\sqrt{(L_m + L_r)C_r}} \quad (10)$$

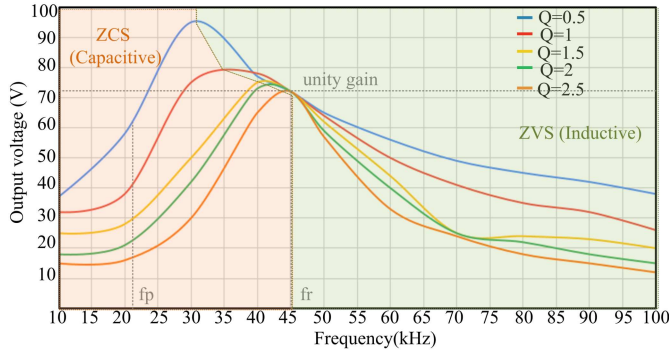


Fig. 3. Gain frequency curve of LLC resonant converter

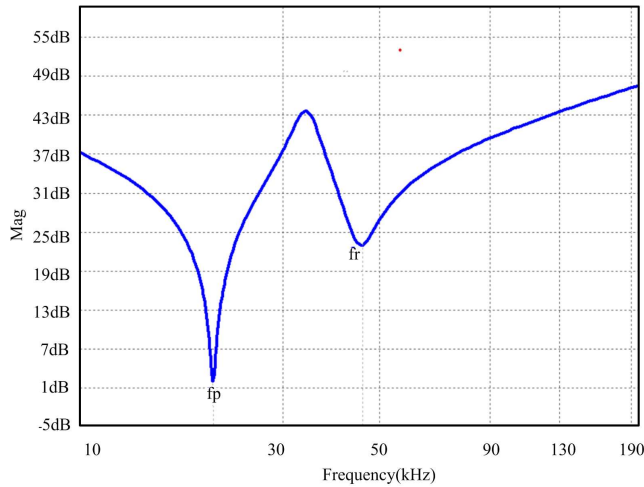


Fig. 4. Impedance Graph of the Resonant Tank

C. LC Filter Design

The ideal value of C_f is given by

$$C_f = \frac{I_{max}}{2f_{line}\pi V_{max}} \tan\theta \quad (11)$$

The filter inductance is given as

$$L_f = \frac{1}{4\pi^2 f_{cor}^2 C_f} \quad (12)$$

where V_{max} and I_{max} is the peak value of the supply voltage and current. f_{line} is the supply frequency which is 50Hz. θ is taken as 1° . The corner frequency is selected as 1/10th of the switching frequency equal to 5kHz.

IV. SIMULATION AND RESULT ANALYSIS

The converter is modeled in MATLAB/SIMULINK to validate the functioning of the converter. The IBPFCC is compared with the conventional Boost PFC converter. The specifications of the converter and designed parameters are specified in Table I. The 110Vrms supply is converted to 310V DC link voltage by the IBPFCC. The duty ratio is chosen as 0.5 to meet the condition specified in section II. The switching frequency of the primary stage and secondary stage converter is chosen as 50kHz. The resonant frequency of the LLC resonant converter is 45kHz. The optimal voltage for charging the battery can be derived when operated at the frequency range of 46kHz to 50kHz. The half-bridge LLC resonant converter gives the unity gain at the resonant frequency, given in Fig 3. The impedance graph of the resonant tank is shown in Fig 4, detailing two resonant frequencies. The maximum gain of the LLC resonant converter is obtained between f_p and f_r . The output voltage is regulated at unity gain for light variations in load when the converter is operated at the resonant frequency. The converter is operated in the ZVS region to reduce the switching losses [11], thus improving the efficiency of the converter, and the ZCS region is avoided to limit the circulating current and use the capacitor energy efficiently [12]. The supply current and voltage of the charger are given in Fig 6. The supply voltage and current are in phase, also the current is sinusoidal, and a power factor of 0.99 is obtained which is satisfactory. The THD of the converter is obtained as 3.87% is provided in Fig 6, and satisfies the standards provided for the battery charger. The supply current THD of the conventional boost PFC converter obtained during comparison with IBPFCC was 9.77%. The inductor currents operated in DCM mode are represented in Fig 7, This interleaving inductor currents significantly reduce the supply current ripples and reduce the inductor peak current compared to the boost converter.

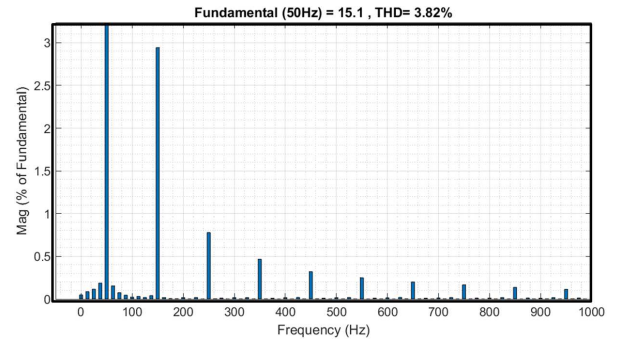


Fig. 5. Supply current THD

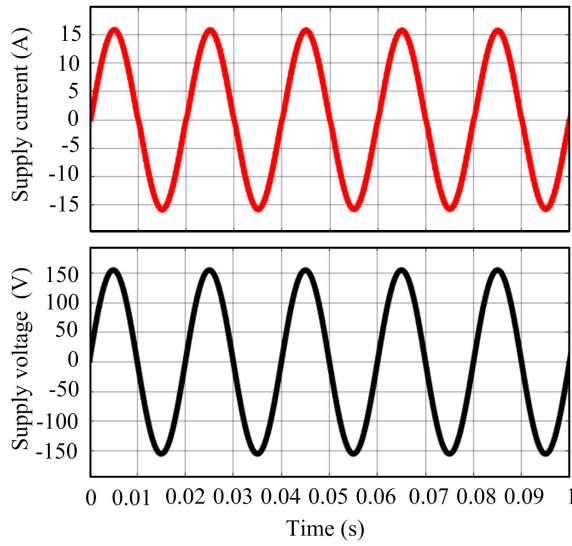


Fig. 6. Supply current and supply voltage

TABLE I
SPECIFICATIONS AND COMPONENTS OF CONVERTER

Selected Parameters	Symbol	Value
Input Voltage	V_{inp}	110 Vrms
DC link Voltage	V_{dclink}	310 V
Power	P_{out}	1kW
Output Voltage	V_{out}	65 V
Output Current Ripple	ΔI_{out}	4%
Designed Parameters	Symbol	Value
Duty Cycle	D	0.5
IBPFCC Inductance	L_{p1} and L_{p2}	90 μH
DC Link Capacitor	C_{p1}	880 μF
Output Filter Capacitor	C_{s1}	40 μF
Resonant Inductance	L_r	74 μH
Magnetizing Inductance	L_m	222 μH
Resonant Capacitor	C_r	169 nF
Input Low Pass Filter Inductance	L_f	330 μH
Input Low Pass Filter Capacitor	C_f	3 μF
Switching Frequency	f_s	50 kHz
Resonant Frequency	f_r	45 kHz
Transformer Turns Ratio	n	2:1:1

TABLE II
COMPARISON OF INDUCTOR CURRENTS BETWEEN PFC CONVERTER AND INTERLEAVED BOOST PFC CONVERTER IN DCM OPERATION

Parameter (Values are in A)	Boost PFC	IBPFCC
RMS Current	11.66	6.18
Average Current	8.09	4.18
Saturation Current	30.90	16.8

The closed-loop operation of the charger achieved by the primary stage converter at varying loads from 1.2kW to 1kW at 2 sec and 1kW to 0.9kW at 4 sec is given in Fig 8. Also, DC link voltage regulation for change in supply voltage supply is given in Fig 8. The supply voltage is varied $\pm 6\%$ of the rated supply voltage, At 1.5 sec a step change in supply voltage from 110V to 102V, and at 3 sec a step change from 102V to 116V was provided. At both these conditions the DC link voltage is regulated to 310V. Also, The diode currents and capacitor currents of the IBPFCC are shown in Fig 10. The inductor

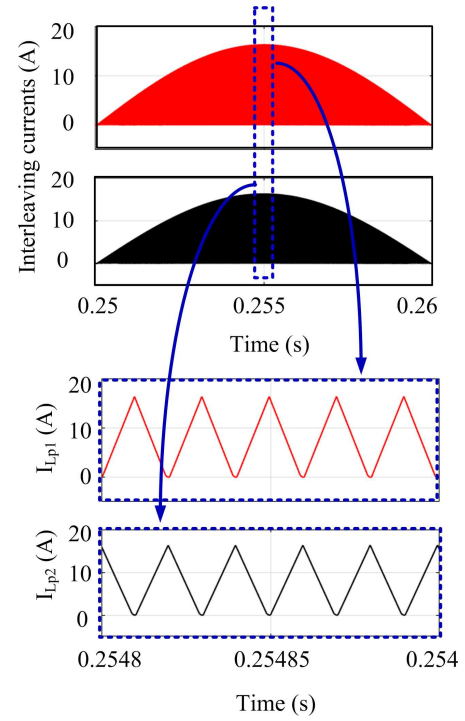


Fig. 7. Inductor currents of IBPFCC

current comparison of the IBPFCC and conventional boost PFC converter is presented in Table II. The magnetizing and resonant inductance current is shown in Fig 11. The output voltage ripple of 2.5% and current ripple of about 6% are observed at the output of the IBPFCC stage. The outputs LLC resonant is given in Fig 12, a ripple of about 1.4% is reflected in the output current.

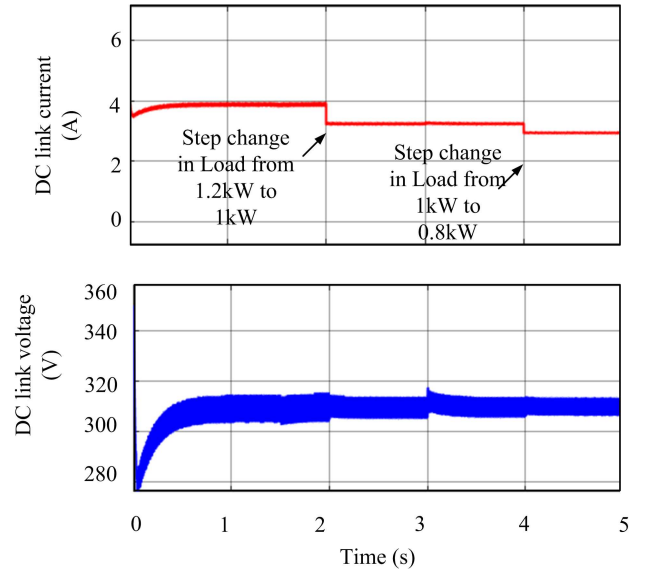


Fig. 8. DC link voltage regulation for variation in load

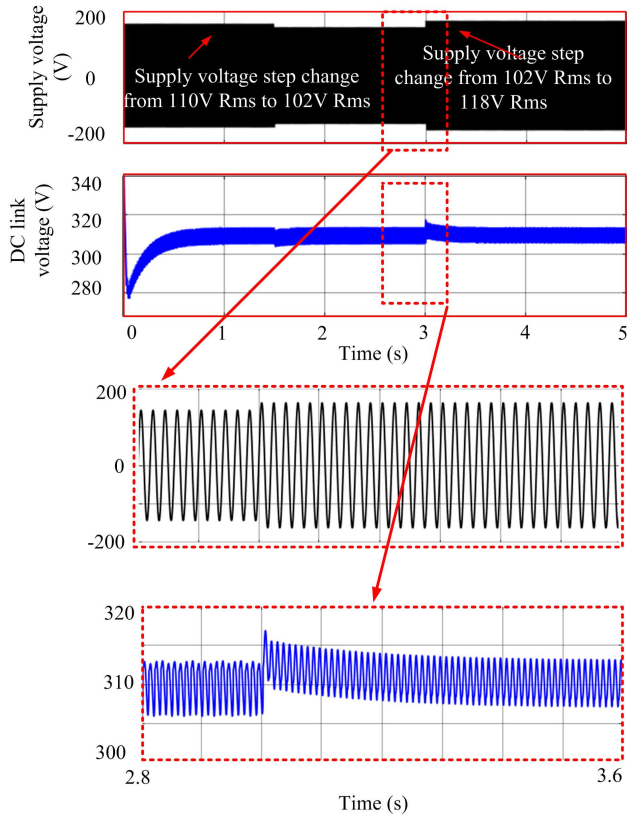


Fig. 9. DC link voltage regulation for variation in supply voltage

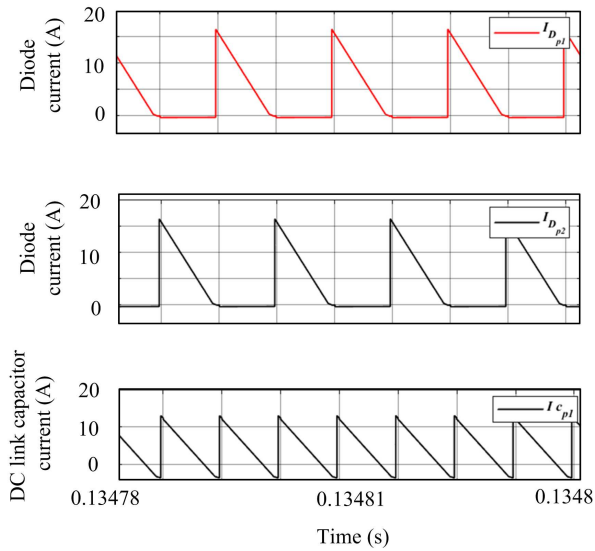


Fig. 10. Diode currents and DC link capacitor currents

V. CONCLUSIONS

This paper considers a two-stage EV battery charger for electric scooters. The primary stage of the charger consists of an interleaved boost PFC converter while the secondary stage consists of a Half Bridge LLC resonant converter. The primary stage converter is operated at DCM providing the advantage

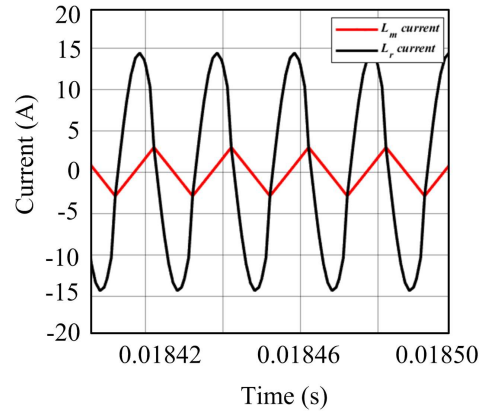


Fig. 11. Resonant tank and magnetizing currents

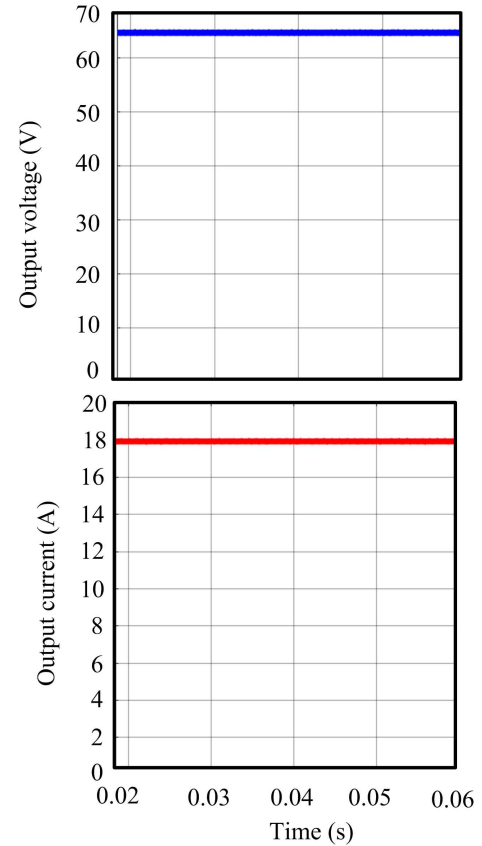


Fig. 12. Output voltage and current of battery charger

of natural power factor correction and reducing the number of sensors required. The secondary stage converter operated in soft switching mode thus reducing the power loss and improving the overall efficiency of the charger. The detailed simulation study of the converter is carried out in MATLAB Simulink. The PFC converter regulates the DC link voltage to 310V with the implementation of a voltage loop control, The LLC Resonant converter, converts the 310V to 65V to charge the battery. The THD and power factor analysis is also given, showing a significant reduction in supply current THD and a

near unity power factor correction. The analysis verifies that the system can perform as per the regulations specified for EV battery chargers.

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